

STRUCTURE OF PREMIXED TURBULENT FLAMES IN THE THIN-REACTION-ZONES REGIME

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In practical combustors, the structure of a turbulent premixed flame has many facets due to the widely varying turbulence-chemistry interactions that can occur. Premixed combustion in the flamelet, the corrugated flamelet, and the distributed reaction (recently called the thin reaction zones) regimes can coexist within the same device. Models using large-eddy simulation (LES) methodology to simulate premixed combustion in practical systems must, therefore, be able to predict these space- and time-varying flame structure and propagation characteristics without requiring ad hoc changes. Here, the linear-eddy model (LEM), developed earlier for the flamelet regime, is used without any modifications to simulate premixed flames over the entire parameter space. A 15-step, 19-species, methane-air mechanism has been used in the *in situ* adaptive tabulation (ISAT) procedure to investigate premixed flame structure from the flamelet to the thin-reaction-zones regime. Qualitative and quantitative comparisons with experimental observations show that the LEM is capable of accurately capturing the flame structure in both flamelet and thin-reaction-zones regimes. These simulations confirm that in the thin-reaction-zones regime the preheat zone thickness increases with increase in the Karlovitz number, Ka . However, the reaction zone remains very thin and of the order of the laminar flame thickness. These results, along with earlier studies of LEM in non-premixed and flamelet premixed combustion, confirm the viability of LEM as a practical combustion subgrid model for use in reacting LES.

Introduction

The structure and propagation characteristics of a turbulent premixed flame depend in large part on the relative magnitude of the chemical and turbulence time and length scales. The ratio of the turbulence intensity, u' , to the laminar flame speed, S_L , and the ratio of the characteristic turbulent scales (e.g., the Kolmogorov scale, η , or the integral length scale, L) to the flame thickness, δ_F , have often been used to characterize the nature of premixed flame-turbulence interactions. Combustion diagrams, such as the one shown in Fig. 1, have been used to characterize the structure of the premixed flame [1,2]. Two regimes are of particular interest here: the flamelet regime, which is characterized by $\delta_F \ll \eta$, where turbulence only wrinkles the flame without affecting its structure, and the distributed reaction regime, characterized by $\eta < \delta_F$, where turbulent eddies can penetrate into the flame, thereby modifying the flame structure.

Studies [3–5] have suggested that even in the distributed reaction regime the reaction zone is very thin and of the order of the laminar flame thickness. This result is derived from the observations that for large $Ka = (\delta_F/\eta)^2$ and Reynolds number $Re = u'L/\nu$, turbulent eddies can enter the preheat zone and, thus, increase turbulent transport of heat and species

away from it. This redistribution can thicken the preheat zone. However, eddies do not penetrate into the reaction zone since they are dissipated by the increased viscous dissipation near the flame. Using these results, the distributed reaction regime has been reclassified as the extended flamelet [3] or the thin-reaction-zones [6] regime. The key implication is that modified versions of flamelet models can be used in the thin-reaction-zones regime, as recently described by Peters [6].

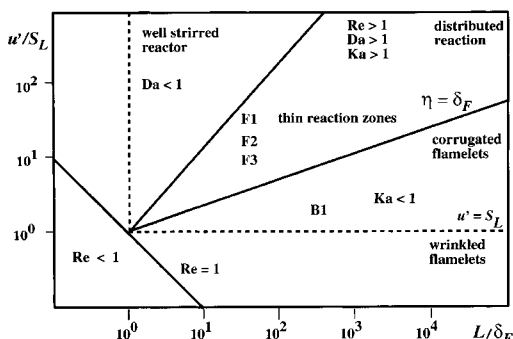


FIG. 1. Diagram of turbulent premixed combustion regimes. The locations of the flames simulated in the present study are also shown.

Simulations of realistic combustion systems using large-eddy simulations (LES) have often employed such flamelet models with reasonable success [7,8]. However, when detailed information of the chemical species distribution is required, flame-turbulence interactions at all the relevant scales must be resolved. In order to do this, a model is needed that can be applied in the entire parameter space without requiring any ad hoc adjustments. In this study, the linear-eddy mixing (LEM) model of Kerstein [9], which was previously used for the flamelet regime [10–12], has been used without any changes for both the flamelet and the thin-reaction-zones regimes. In particular, we simulated the methane-air flames B1, F3, F2, and F1 that were recently studied [4,5,13]. The typical locations of these flames are given in Fig. 1. As shown, B1 is in the corrugated flamelet regime, while the flames F3, F2, and F1 cover the entire regime in the distributed regime.

Model Formulation

The details of the LEM model have been given elsewhere [9–12] and, therefore, are only briefly summarized here. LEM is a stochastic model which treats reaction-diffusion and turbulent convection separately but concurrently. Reaction-diffusion processes evolve on a one-dimensional domain in which all the characteristic length scales in the turbulent field (from L to η) are fully resolved. The orientation of the one-dimensional domain is in the direction of the scalar gradient [9], and within this domain, the equations governing constant pressure adiabatic laminar flame propagation are

$$\frac{\partial Y_k}{\partial t} = -\frac{1}{\rho} \frac{\partial \rho Y_k V_k}{\partial s} + \frac{\dot{\omega}_k W_k}{\rho} + F_{k\text{stir}} \quad (1)$$

$$\frac{\partial T}{\partial t} = -\frac{1}{\rho c_p} \sum_{k=1}^N c_{p,k} Y_k V_k \frac{\partial T}{\partial s} + \frac{1}{\rho \bar{c}_p} \frac{\partial}{\partial s} \left(\bar{\kappa} \frac{\partial T}{\partial s} \right) - \frac{1}{\rho \bar{c}_p} \sum_{k=1}^N h_k \dot{\omega}_k W_k + F_{T\text{stir}} \quad (2)$$

The equation of state for the scalar mixture is $P = \rho T \sum_{k=1}^N Y_k R_u / W_k$, and the caloric relation is given by $h_k = \Delta h_{f,k}^0 + \int_{T^0}^T c_{p,k}(T') dT'$. Here, T is the temperature, P is the thermodynamic pressure, R_u is the universal gas constant, and ρ is the mass density. Also, Y_k , W_k , $c_{p,k}$, $\dot{\omega}_k$, h_k , V_k , and $\Delta h_{f,k}^0$ are, respectively, mass fraction, molecular weight, specific heat at constant pressure, mass reaction rate, enthalpy, diffusion velocity, and standard heat of formation (at standard temperature, T^0) of the k th species. The mixture-averaged specific heat at constant pressure and thermal conductivity are, respectively, $\bar{c}_p$ and $\bar{\kappa}$. The diffusion velocity is given by Fick's law: $V_k = -D_k/Y_k(dY_k/ds)$, where D_k is the mixture averaged diffusivity of the k th species.

Since turbulent convection is implemented explicitly, the convective terms $u \partial Y_k / \partial s$ and $u \partial T / \partial s$ in the species and energy equations are symbolically represented as $F_{k\text{stir}}$ and $F_{T\text{stir}}$, respectively. These terms are implemented using stochastic rearrangement events called triplet maps, each of which represents the action of a turbulent eddy on the scalar fields. It has been shown [9] that this mapping can capture correctly the physical increase in scalar gradient (without affecting the mean scalar concentration) due to eddy motion. Three parameters are needed to implement these turbulent stirring events: eddy size, l , the eddy location within the one-dimensional domain, and the stirring frequency (event rate), λ . The eddy size in the range L to η is determined randomly from an eddy size distribution, $f(l)$, obtained using inertial range scaling in three-dimensional turbulence [9]: $f(l) = (5/3)l^{-8/3}/(\eta^{-5/3} - L^{-5/3})$. Here, η is determined from the inertial range scaling law $\eta = N_\eta L Re^{-3/4}$, where N_η is an empirical constant. This constant reduces the effective range of scales between L and η but does not change the turbulent diffusivity, as described earlier [11].

The event location is randomly chosen from a uniform distribution, and the event rate (frequency per unit length) is determined by [9]

$$\lambda = \frac{54}{5} \frac{\nu Re}{C_\lambda L^3} \frac{[(L/\eta)^{5/3} - 1]}{[1 - (\eta/L)^{4/3}]} \quad (3)$$

The time interval between events is then given as $\Delta t_{\text{stir}} = 1/(\lambda X_{\text{LEM}})$, where X_{LEM} is the length of the one-dimensional domain.

The unique feature of the LEM model is that although the scalar evolution is simulated in one dimension, the effect of turbulence on the scalar fields is modeled using three-dimensional scaling laws. As a result, flame wrinkling occurs at spatial and temporal scales that mimic effects of realistic three-dimensional turbulent eddies on the laminar flamelets. The ability of LEM to capture realistic flames in the flamelet regime is now well documented [11,12].

The above formulation has two constants, C_λ and N_η , both of which arise from the use of scaling laws. In the earlier studies [11,12], these parameters were obtained by comparing LEM predictions to experimental data [14] in the flamelet regime. The present study uses these same values ($C_\lambda = 15$ and $N_\eta = 4$) in order to determine if the LEM model validated in the flamelet regime can be applied to flames in the thin-reaction-zones regime without any adjustments.

Numerical Implementation

Methane-air flames similar to those in the experiments were studied. In order to obtain a realistic chemical state over a wide range of operating conditions, a 15-step, 19-reaction skeletal mechanism

TABLE 1
Turbulent flame properties

Flame	ϕ	Re	Da	Ka	S_L (m/s)	δ_F (mm)	u' (m/s)	l (mm)	u'/S_L	l/δ_F
B1	0.8	496	37.08	0.8175	0.30	0.233	0.52	15.0	1.733	64.28
F3	1.0	73	2.581	23	0.40	0.175	2.12	2.4	5.313	13.71
F2	1.0	125	1.511	63	0.40	0.175	3.63	2.4	9.075	13.71
F1	1.0	164	1.152	91	0.40	0.175	4.76	2.4	11.90	13.71

[15] was employed. This mechanism (which included NO_x kinetics) has been shown to be quite accurate over a wide range of equivalence ratios. It is also capable of predicting extinction and reignition, which is particularly relevant here since flame F1 is considered close to the extinction limit [5].

The numerical method is the same as in the earlier study [11,12] and, therefore, only briefly summarized. To simulate a stationary flame, a moving observation window was used that translates with the flame brush to maintain approximately the same relative position between flame center and observation window (even though the flame propagates freely into the reactants). All statistics are obtained relative to the flame center. The computational domain is chosen large enough (typically $6L$) to fully capture the flame brush. Earlier studies [11,12] and the present study showed that statistically, stationary flames can be simulated using this approach. Further details have been given elsewhere [11,12].

All spatial derivatives in equations 1 and 2 were discretized using a second-order accurate central difference scheme. A fractional operator splitting method [11,16] was used to first evolve (via molecular diffusion and turbulent stirring) the scalar equations (without the reaction terms) for a small time step, δt , to obtain the thermochemical state after mixing. Then, the reaction system was integrated over δt to get the final scalar state. The reaction point problem is very stiff due to a wide range of characteristic chemical time scales, and hence, the computational cost for the time integration can be very high. Here, the *in situ* adaptive tabulation (ISAT) approach [15] was used to efficiently handle the finite rate kinetics [17].

In ISAT, only the accessed region of the composition space, which is a subset of the whole realizable region (the set of all possible combinations of compositions for a given number of species), is tabulated. This tabulation is done as a part of the simulation, and when the same composition re-occurs, the table is searched and the stored information is retrieved using a fast binary tree search algorithm. Since only the accessible region is stored, the overall time required to build, retrieve, and store information is significantly reduced. Further details of the ISAT algorithm have been given elsewhere [17] and, therefore, are avoided here for brevity.

To further reduce computational cost, the present simulation code was implemented on parallel systems using message passing interface (MPI). For the chemistry point problem, no communication was needed. Each processor built a table for the composition that occurred inside its domain during the computation. This localized the ISAT table to each processor and reduced the overall load (including search and retrieve time) for each processor (as opposed to building a single table for all the composition that occurs over the whole of the domain). All simulations were done on a 32-processor CRAY-T3E system.

Computational efficiency of ISAT is significant, as reported earlier [17]. The size of the table and the cost depend on the error tolerance (which determines the allowable error in each of the scalars for a given initial state). For high accuracy, this parameter should be low, but this can lead to a significant increase in the total simulation time and storage. For a value of $\epsilon_{\text{tol}} = 0.0008$, a speedup of around 30 in the chemical update is achieved by using ISAT. This is consistent with the speedup reported earlier [17].

Results and Discussion

Instantaneous Flame Structure

The LEM simulations were carried out long enough to obtain sufficient data for statistical analysis. Instantaneous profiles of the temperature and the destruction rate of methane, $\dot{\omega}_{\text{CH}_4}$, for flames B1, F3, and F1, are shown in Fig. 2a, 2b, and 2c, respectively. We use $\dot{\omega}_{\text{CH}_4}$ to identify the location of the reaction zone (CH is not available in the skeletal mechanism) since it has been shown to correlate well with the location of the reaction zone [18]. The transition from the flamelet (flame B1) to the structure seen in the thin-reaction-zones regime is apparent by comparing these figures. For flame B1, the preheat zone upstream of the flame zone is inert, and its temperature is close to the free stream value. However, as Ka and Re increase (flames F3 and F1), eddies penetrate into the preheat zone and increase the temperature ahead of the flame brush. Progressing from the conditions for flame F3 to those for flame F1, the temperature increase in the preheat

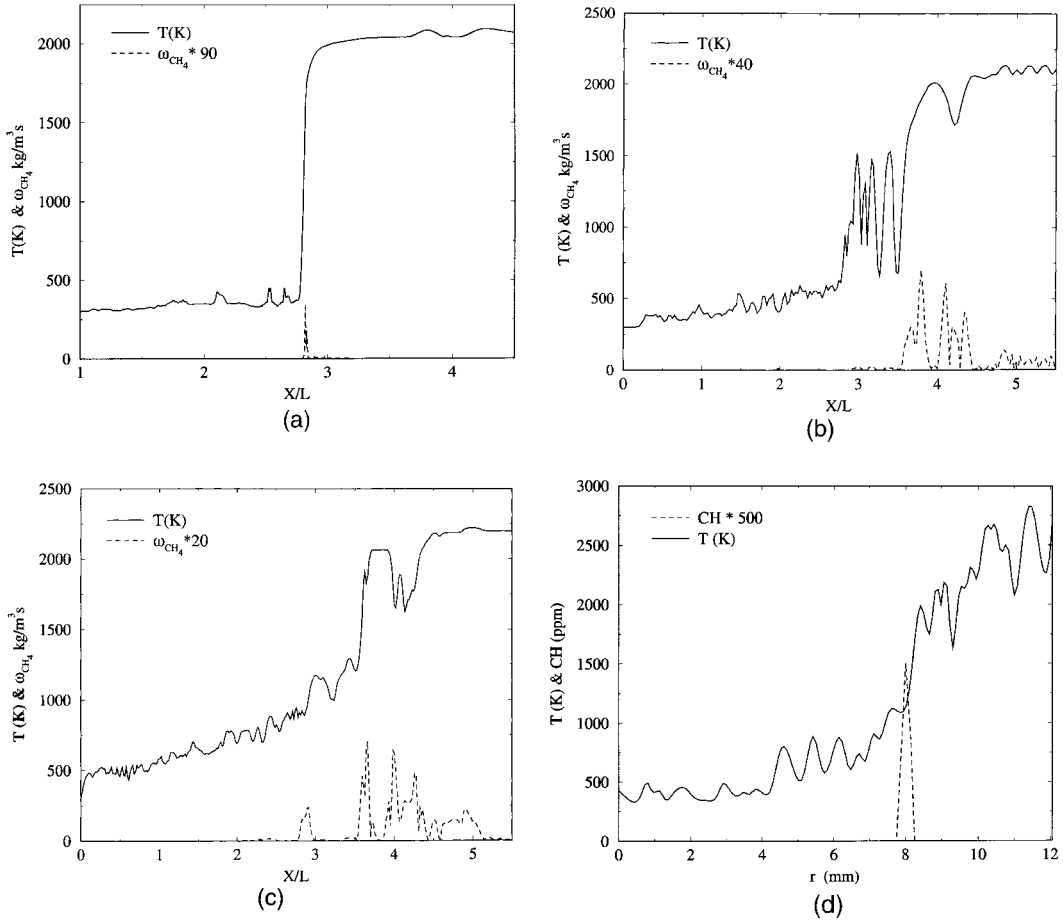


FIG. 2. Instantaneous temperature and ω_{CH_4} profiles for the flames. Note that the scale of ω_{CH_4} is changed for ease of presentation. (a) Flame B1, (b) flame F3, (c) flame F1 (LEM), (d) flame F1 (experiment).

zone increases. However, the reaction zone still remains localized and thin in all cases, with ω_{CH_4} peaking at the location of a steep temperature gradient. A very good qualitative agreement of the temperature profile (for the flame F1) with the experimental data is observed, as can be seen by comparing Fig. 2c and 2d (from Mansour et al. [5]). The reaction zone thickness (using the definition given in Ref. [5]) is estimated to be 0.16 mm, 0.21 mm, and 0.27 mm for F1, F2, and F3 flames, respectively. These values are very close to the laminar flame thickness of 0.175 mm [4]. Experimental data [5] for CH suggests a value in the range 0.2–0.5 mm for the F3 and F1 flames. Thus, the numerical predictions are within the measured experimental scatter.

Although the reaction zone is thin and of the order of the laminar flame thickness, the preheat zone is much larger. Using the definition [5] that the preheat zone is between 600 K and 1300 K, the thickness of the preheat zone is estimated as 2.2 mm, 1.92 mm,

and 1.92 mm for the flames F1, F2, and F3, respectively. In recent experiments [5], the preheat zone thickness was estimated in the range 2–3.5 mm for flame F1 (no other data is currently available). Thin reaction zone, with a thick preheat zone in front, is characteristic of flames in thin-reaction-zone regime, and the present simulations appear to have accurately captured the characteristics of the flame structure in this regime.

Mean Flame Structure

Mean and variance of the reaction progress variable, $c = (T - T_u)/(T_b - T_u)$, where T_u and T_b are, respectively, the reactant and product temperature, have also been computed. The transition from flamelet to thin-reaction-zones regime combustion is best characterized by the probability density function (PDF) of c . Fig. 3 shows the PDFs at six different locations for the flames B1, F3, F2, and F1,

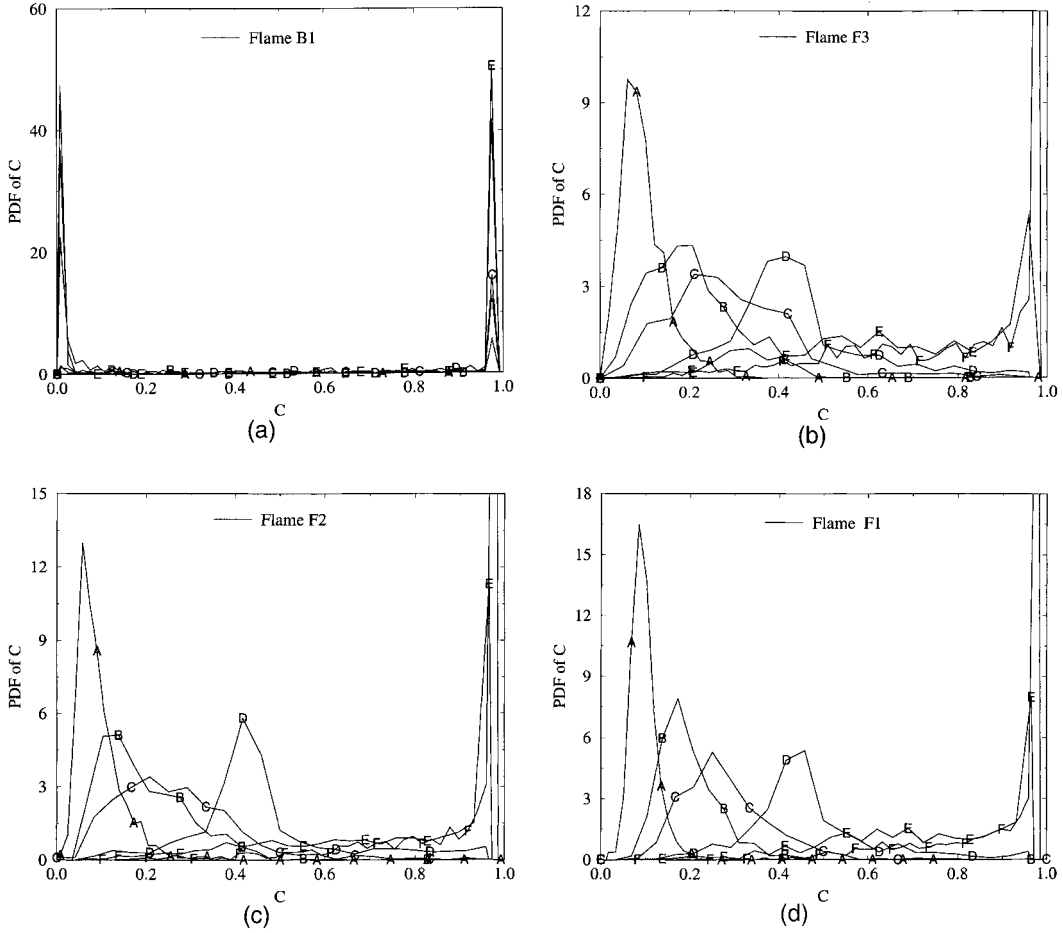


FIG. 3. PDF of progress variables, c , at six different locations in the flames. The transition from flamelet structure (bimodal PDF for flame B1) to the thin-reaction-zones flame structure is apparent in these PDFs. A, $c = 0.1$; B, $c = 0.2$; C, $c = 0.3$; D, $c = 0.5$; E, $c = 0.8$, F, $c = 0.9$.

respectively. As expected, in the flamelet regime, the PDF exhibits two peaks corresponding to unburned and burned gases, respectively, since in this regime probability of the reactive state corresponding to the transition between reactants and products is small.

For flames F3, F2, and F1, which are in the thin-reaction-zones regime, the PDF is not entirely bimodal. The small scales of turbulence modify the transport within the reactive states. Small eddies that enter the preheat zone broaden the flame, increasing the probability of intermediate values of the progress variable. There is excellent qualitative agreement with the progress variable PDFs reported in Ref. [19] for flames in similar regimes.

The mean temperature profiles for flames B1, F3, and F1 are shown in Fig. 4. A direct comparison shows that as Ka increases, the flame brush thickness increases, which results in a broadened mean tem-

perature profile. An increase in preheat zone thickness can be observed as we move from flamelet to distributed regime. The thickness of the preheat zone is of the order of the integral length scale for all the F flames. This suggests that eddies as large as L are involved in the transport of heat and mass from the reaction zone into the preheat zone.

Scalar dissipation rates in the flames are compared by observing the maximum scalar variances in them. Variance of the progress variable for flames B1, F3, and F1 is shown in Fig. 5. Results show that flame F1 has a lower maximum scalar variance when compared to other flames. Since flame F1 has a higher Reynold's number compared to flame F3, it has more small-scale wrinkling near the inner layer than does F3. This increases the fuel dissipation rate and decreases the scalar variance of flame F1 when compared to F3. This is consistent with the increased

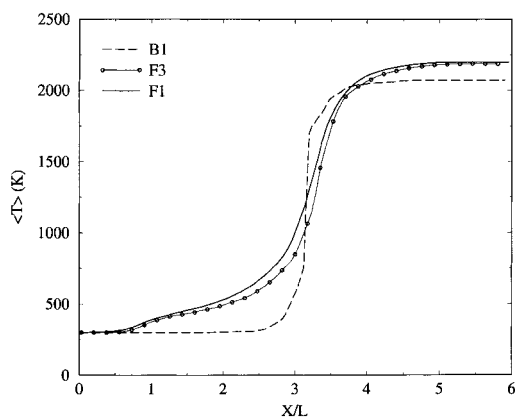


FIG. 4. Mean temperature profile across the flame brush for flames B1, F3, and F1. Note that, the integral length scale, L , is different for flame B1 (Table 1).

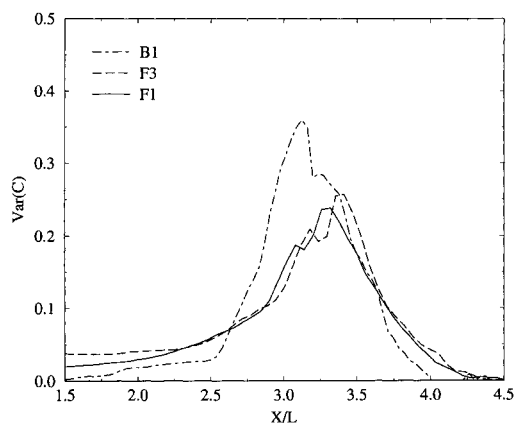


FIG. 5. Scalar variance profiles for flames B1, F3, and F1.

scalar dissipation rate observed for flame F1 compared to flame F3 in the experiments of Chen et al. [20].

The mean profiles of the major and minor species are also available from the LEM simulation and have been analyzed, but relevant discussion is deferred to the future.

Turbulent Propagation Speed

Turbulent propagation speed can also be determined from the time trace of the propagation of the leading flame [12]. A u_t/S_L of 4.7 was reported for flame B1 [13], while the current LEM prediction is 4.81. This is considered to be a reasonable agreement. Estimation of the turbulent flame speed in the thin-reaction-zones regime is problematic, but an estimation approach was proposed earlier [6] using the

G-equation model. Using this approach, u_t/S_L is predicted to be 4.05, 6.35, 7.53, and 9.95, while the corresponding LEM simulated predictions are 4.81, 4.65, 5.96, and 7.84 for flames B1, F3, F2, and F1, respectively. Although the actual numbers are different, the increase in u_t/S_L with increase in Ka is captured in both predictions. Since a steady-state burning velocity cannot be defined without ambiguity in the thin-reaction-zone regime, the value of u_t/S_L reported for flames F2 and F1 may not have any physical meaning.

Conclusion

Structure of the premixed turbulent flames in various combustion regimes has been investigated using a scalar mixing model. The flames simulated closely corresponded to flames studied in recent experiments and range from the flamelet regime to the (estimated) end of the thin-reaction-zones regime. The LEM model, validated earlier for the flamelet regime, was used without any modification to study all these flames. A skeletal methane-air mechanism was used within an ISAT algorithm to significantly reduce the computational effort. Results show reasonable agreement with past experimental observations and confirm the flamelet nature of the thin-reaction-zones regime. The LEM model appears capable of capturing quantitatively the structure and propagation characteristics of turbulent premixed flames over a wide range of operational parameters without requiring any ad hoc adjustments. The computational efficiency gained by using the ISAT procedure also provides the confidence that the LEM model with skeletal kinetics is a good candidate for a subgrid combustion model for large-eddy simulation (LES) applications. This is particularly important since turbulence and combustion conditions can vary widely within the same device, and the simulation model must be capable of dealing with these regimes without requiring ad hoc fixes. The past use of LEM for subgrid, non-premixed [21] and premixed flamelet combustion [12,22] and the present study confirm that LEM is a viable subgrid combustion model regardless of the nature of combustion. Application of LEM within LES using the ISAT procedure is currently underway and will be reported in the near future.

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COMMENTS

Friedrich Dinkelacker, University of Erlangen, Germany. You used your linear-eddy model to calculate properties and structure of highly turbulent premixed flames in a remarkable way. If I understood you right, these calculations are one-dimensional. Is there the intention and the possibility to extend this model to two or three dimensions? Or is the general modeling strategy such that the detailed results of the LEM calculations might be used as input to more conventional three-dimensional calculations using, for example, pdf shapes?

Author's Reply. The basic idea of LEM is to represent the small-scale mixing effects by explicitly distinguishing the different processes such as turbulent mixing, molecular mixing, and chemical reaction at all scales. This can be achieved by a reduced one-dimensional description of the scalar field, which makes it affordable to resolve all length scales of the scalar field. So, the general modeling strategy is to use a reduced one-dimensional model at the subgrid level, which will provide input to the conventional three-dimensional LES. Even though the simulation at the subgrid level is one-dimensional, the effect of turbulence on the scalar fields is modeled using three-dimensional inertial range scaling laws.

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Josette Bellan, Jet Propulsion Laboratory, USA. Is there any theoretical justification for the triplet map? And if there is not, then what would happen if you increased the linear dimension by another factor, rather than by 3? If the triplet map is heuristic, given the limited set of results that you presented, isn't it premature to suggest that LEM + ISAT is a viable SGS model for utilization in LES?

Author's Reply. The ability and the accuracy of LEM has been demonstrated in numerous studies (some of which are referred in this paper). The superior ability of LEM as a subgrid model in LES has also been shown in non-premixed combustion studies of Calhoon [Ref. 16 in paper] and premixed combustion studies of Chakravarthy [Refs. 21,22 in paper]. These studies employed a global one-step chemistry of the G-equation. The inclusion of detailed chemistry with ISAT approach is considered a natural extension to deal with more complex kinetics and to capture the finer details of the flame structure as demonstrated in this paper.

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Norbert Peters, RWTH Aachen, Germany. In the thin reaction zone, regime flame curvature has been identified as the dominant propagation mechanism. This is a multi-dimensional effect, which could, however, be included in the one-dimensional formulation of the LEM equation as an additional curvature term. Have you considered to include such a term?

Author's Reply. It is not possible to explicitly include curvature effects in the one-dimensional LEM when the reaction-diffusion equations are solved. It is, however, feasible if the G-equation is used. However, in the LEM the effect of turbulence on the scalar fields is modeled using three-dimensional inertial range scaling laws. As a result, flame wrinkling and the curvature effects occur at spatial scales that mimic the effect of realistic three-dimensional turbulent eddies. This is evident in the present study since the propagation of flames in thin reaction zone regime has been captured with LEM.